KARTIKYA MOTWANI

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WORK EXPERIENCE

Full-Stack AI Developer (Project) | Ergosphere Solutions *Apr – May 2026*

- Built BookIQ, a full-stack AI book platform, end-to-end in 48 hours — Django REST backend, React.js frontend, ChromaDB vector DB, and Gemini AI — deployed live on Railway
- Completed 19/19 assigned tasks: JWT auth, RAG pipeline, rate limiting, Docker deployment, automated testing (8/8 passing), health check endpoint, and GitHub Wiki documentation
- Evaluated RAG accuracy using nomic-embed-text-v1 — achieved MRR@3: 1.000 and 100% Hit Rate vs ChromaDB default (~70%), documented in Jupyter notebook
- Live: book-intelligence-platform-production.up.railway.app | GitHub: github.com/Kartik-061/book-intelligence-platform

Python Developer Intern | Good Dopamine Technologies *Mar – Apr 2026*

- Scraped, cleaned, and verified 4,000+ US university B2B contacts across 3,500+ institutions targeting 4 decision-maker roles per university, achieving ~1% bounce rate via NeverBounce
- Built parallel 1,000+ contact Indian university database (IITs, IISERs, NITs) by cross-referencing official .ac.in/SRIC pages to handle role rotation every 1–2 years
- Automated full data pipeline (dedup, formatting, email merge) using Python + openpyxl, cutting manual processing time by ~70%
- Identified 'Head of Libraries' as highest-ROI bulk-send segment; launched 4 role-specific email templates reducing campaign setup time significantly

Cyber Security Intern | Bleep Education LLP × E-cell IIT Bombay *Jan – Apr 2026*

- Completed 4-month intensive program covering network security, ethical hacking, and information security — earned 3 certificates (Experience, Internship, Training)

Python Developer (Freelance) | Fiverr & Upwork *Nov 2025 – Present*

- Building Selenium-based browser automation with human-mimic algorithms (randomized delays, mouse pathing) for clients; offering web scraping and AI integration services
- Contributing as developer to Sneaker Horizon, an Indian sneaker marketplace startup (brother's venture)

PROJECTS

Alpha Terminal – NSE Algorithmic Trading Platform | *Aug 2026*

Stack: Python · FastAPI · PostgreSQL (Neon) · React.js · SQLAlchemy · JWT Auth · Angel One SmartAPI

- Built a full-stack algorithmic trading platform: near-real-time price feed via Angel One SmartAPI, multi-user JWT auth, and per-user bot isolation via a custom BotRunner/UserBotManager architecture
- Implemented rigorous statistical significance testing (z-test) pooling trade P&L across 15 NSE large-cap stocks; only mean_reversion passed at p=0.0008, later re-validated at p=0.0344 across an expanded 43-stock watchlist — 3 other strategies honestly reported as not significant
- Modeled real Indian trading costs (brokerage capped at ₹20, STT, GST, stamp duty) instead of assuming zero-cost trades; capital-percentage position sizing with max-5-concurrent-position cap
- Currently paper trading live on NSE across a 43-stock watchlist, with automated crash recovery and portfolio state that persists across restarts and redeploys
- GitHub: github.com/Kartik-061/trading-bot

BookIQ – AI-Powered Book Intelligence Platform | *Apr – May 2026*

Stack: Python · Django REST · React.js · ChromaDB · Gemini AI · Docker · Railway

- 588+ books scraped from Open Library & Gutenberg; semantic search via ChromaDB vector embeddings; RAG-based Q&A using Gemini 2.5 Flash
- Full auth system (JWT register/login/logout), protected routes, thumbs up/down feedback loop, rate limiting (30 req/min anon, 60 req/min user)
- 8/8 automated tests passing; Docker multi-stage build deployed on Railway; UptimeRobot monitoring for 100% uptime

Lost & Found Web Application | *Nov 2025*

Stack: Python · Flask · SQLite · HTML/CSS

- Built full-stack campus lost & found portal with user authentication, item listing, search, and responsive UI — solo project from scratch

AI Audio Assistant — Team 'Error 404' | *Dec 2025*

Stack: Python · Audio Processing Libraries

- Collaborated with 4-member team to build voice-activated AI assistant for Bennett University's Computational Thinking course; owned AI logic and project documentation

TECHNICAL SKILLS

Languages: Python (Advanced), Java, C++, JavaScript, HTML/CSS, Verilog

Backend / APIs: Django REST Framework, FastAPI, Flask, REST APIs, SQLAlchemy, PostgreSQL

Frontend: React.js, Tailwind CSS

AI / Data: Gemini API, ChromaDB, RAG Pipelines, nomic-embed-text, Prompt Engineering, MATLAB

Automation: Selenium, BeautifulSoup, openpyxl, Pandas, Web Scraping

DevOps & Tools: Git, GitHub, Docker, Railway, Linux, VS Code, Postman

Cybersecurity: Network Security, Ethical Hacking (fundamentals), Information Security, TryHackMe Pre-Security (Jul 2026)

EDUCATION

B.Tech – Computer Science & Engineering (Cybersecurity Specialization) *2025 – 2029*

Bennett University, Greater Noida • SGPA: 9.0 | 50% Merit Scholarship

Senior Secondary (XII) – Science *2024*

Indra International School, Kota (CBSE) • 80% | 20% Scholarship at ALLEN

Secondary (X) *2022*

Sanskriti The School, Ajmer (CBSE) • 82%

CERTIFICATIONS

- Cyber Security Internship & Training — Bleep Education LLP × E-cell IIT Bombay (Apr 2026)
- Digital Electronics · Sequential Digital Circuits · Combinational Digital Circuits · Electronic Circuits & Number System — Infosys Springboard (Feb 2026)
- Machine Learning with MATLAB — MathWorks (Sep 2025)
- Introduction to Python — Infosys (Nov 2025)
- SmartBU Hackathon-2025 — Bennett University (Sep 2025)
- TryHackMe Pre-Security Learning Path — TryHackMe (Jul 2026)